

BOX HW4

Minimal information for a node-centric de Bruijn graph

For a node-centric de Bruijn graph of order k , the minimal information needed is the set of k -mers, which are the vertices. Edges do not need to be stored, they can be recomputed at any moment.

Storing only canonical k -mers accounts for both strands. One entry thus represents both strands, which halves the memory needed.

A dictionary maps canonical k -mers to their occurrence counts. The keys are still the vertices, so no information is lost, and the multiplicities are needed later (using Python's `Counter`).

Multiplicity of the k -mers

Fasta file	k -mers processed	Distinct k -mers
Perfect reads, forward	2100000	149 866
Perfect reads, both strands	2100000	149 869
0.1% errors	2100000	213 031
1% errors	2100000	702 101

The multiplicity distribution has two parts. The perfect datasets show a single bell centered around 13-14, and a thin tail beyond 30 corresponding to repeated regions of the genome. The erroneous datasets add a large peak at multiplicity 1, made of k -mers created by sequencing errors. At 1%, the bell is also shifted to the left.

Threshold t values

On the erroneous reads, most k -mers seen only once are sequencing errors: the multiplicity histogram shows a peak at 1, clearly separated from the bell of true k -mers (centered around 13-14). Using the graph of the perfect reads as ground truth, false positives (erroneous k -mers kept) drop from 552 239 at $t=1$ to 8 219 at $t=2$ and 20 at $t=3$ for the 1% dataset (63 163, 76 and 0 for 0.1%), while false negatives (true k -mers lost) grow by a factor of about 2 to 4.5 per step.

I choose $t=3$, which minimizes false positives + false negatives on both datasets (56 for 0.1%, 533 for 1%). At $t=4$ all false positives are gone but three times more true k -mers are lost (1% dataset), whereas at $t=2$ thousands of errors remain. The 20 remaining false positives at $t=3$ (1% dataset) are likely errors repeated at least three times, which no threshold can separate from true k -mers.

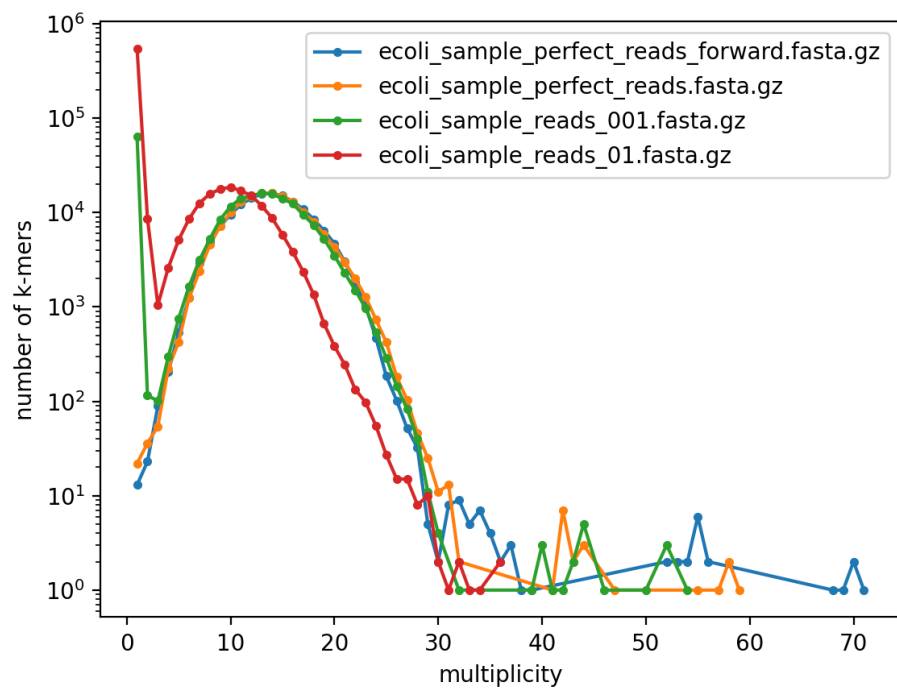


Figure 1: Multiplicity of k-mers for each files

Recovering unitig

Algorithm, some principles

- Two way search: The function builds the unitig by exploring the graph to the right and to the left
- Stopping at branches: It stops when a node has more than one successor, or if the successor has more than one predecessor
- Avoiding cycles: It stops if it finds a k-mer that was already visited, to prevent cycles

Results

Size (in bp) of the unitig containing the given 31-mer.

Dataset	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Perfect reads, forward only	5 586	5 575	5 574	5 567
Perfect reads, both strands	5 588	5 581	3 243	3 243
0.1% errors	46	4 811	5 581	5 332
1% errors	31	757	3 095	2 243

On the perfect datasets, the unitig is nearly identical for both at $t=1$ and 2. It covers only about 4% of the 150 kbp region. Without errors, this likely indicates a repeat-induced fork. It decreases when t increases, because true k-mers of low multiplicity are removed, and each removal cuts the path.

Erroneous datasets exhibit two effects. At low t , erroneous k-mers create forks that stop the unitig almost immediately. When t increases, the false positives disappear and the unitig grows: 5 581 bp at 0.1% for $t=3$, identical to the perfect reads at $t=2$! The size peaks at $t=3$. However, a single missing k-mer is enough to cut the unitig, so this size depends on the sample of reads. With 1% errors, about 27% of k-mers contain an error (against 3% at 0.1%), so the unitig stays shorter for every t than at 0.1% and never exceeds 3 095 bp.

Sum of the size of the recovered unitigs

Dataset	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Perfect reads, forward	150 496	150 483	150 490	150 701
Perfect reads, both strands	150 499	150 477	150 531	150 687
0.1% errors	419 101	150 888	150 683	150 822
1% errors	2 233 421	194 095	152 526	155 321

The total size for perfect reads remains approximately 150.5kbp across all thresholds. At $t = 1$ the datasets with errors give much larger totals. Erroneous

k-mers form thousands of short, overlapping unitigs. The total drops at $t=2$ and is minimal at $t=3$ for both datasets, which confirms (again) the choice of Q7. It increases again at $t=4$ because true k-mers with low multiplicity are removed.